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Batch: 9:30-11:30

Date: 03/10/26

① What is static data and a static method? Explain with code.

⇒ It is shared data and method which is ~~not~~ belong to ~~class~~

⇒ Static methods can't access Non Static datas.

Example:-

```
import java.io.*;
```

```
Static int a;
```

```
class Sample {
```

```
    static int a = 10;
```

```
    static void display() {
```

```
        Sop(a);
```

```
    }
```

② Exception:-

Exception is seen time even

If it occurs it interrupts the normal flow of program.

Example:-

```
import java.io.*;
```

```
import util java.util.*;
```

```
class SampleException extends Exception {
```

```
    public SampleException (String m)
```

```
    {
```

```
class test {
```

```
    public static void main (String[] args)
```

```
    {
```

```
        Scanner s = new Scanner (System.in);
```

```
        System.out.println ("Enter your  
                                name");
```

```
        String name = s.next();
```

```
    try {
```

```
        if (name.matches ("[a-zA-Z]+")) {
```

```
            Sop ("proceed the code");
```

```
        }
```

```
    else {
```

```
        SampleException ex = new Sample  
                                Exception ("
```

```
                                throw ex;                                Not support")
```

```
        }
```

```
    }
```

```
    catch (SampleException ex)
```

```
    {
```

```
        Sop (ex.getMessage());
```

```
    }
```

```
}
```

```
}
```

③ Armstrong Number

```
import java.io.*;
import java.util.*;

class ASN {
    public static void main (String[] args) {
        Scanner s = new Scanner (System.in);
        SOP("Enter your No");
        int n = s.nextInt();
```

```
        int r = 0;
        int s = 0;
```

```
        while (n > 0) {
```

```
            {
```

```
                r = n % 10;
```

```
                s = s * 10 + r;
```

```
            } if (n > 0) {
```

```
                s = s * 10;
```

```
            }
```

```
        } System.out.println(s);
```

```
    }
```

```
}
```

123
10
if (r < 9)
r = r * 10 + s

④ Polymorphism:-

Polymorphism is same name with different performances.

Types :-

⇒ Static polymorphism
(runtime)

⇒ dynamic (compile time)

(i) Static polymorphism-

It is runtime polymorphism because it will know that static Polymorphism at runtime.

* Example

=> Method overloading

~~class~~

```
public void addition (int x, int y) {
```

```
    int z = x + y;
```

```
    Sop (z);
```

```
}
```

```
public void addition (int x, float y) {
```

```
    int z = x + y;
```

```
    Sop (z);
```

```
}
```

(ii) Dynamic Polymorphism:-

It is compile time because it will override the method it will know at the compile time if we didn't override any method it will show error.

Example:-

Method override.

```
class Sample {
```

```
    public void circle ();
```

```
    public void square ();
```

```
}
```

```
class Sample1 {
```

```
    public void circle () {
```

```
        Sop ("circle"); }
```

```
    public void square () {
```

```
        Sop ("square"); }
```

```
}
```